TUTORIAL 11

Steps

1. Open PyCharm and create a new project named SoftwareTesting.
2. Right-click the project → New → Python File. Name it tutorial11.py.
3. Open the PyCharm Terminal and install Selenium by running: pip install selenium
4. Copy the complete Python code below into tutorial11.py.
5. Save the file (Ctrl + S), right-click it, and select Run 'tutorial11'.
6. Chrome opens automatically. The program logs in, adds a product, opens the cart, and proceeds to checkout.
7. Check the PyCharm console for PASS messages.

Python Program

```
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC

driver = webdriver.Chrome()
driver.maximize_window()
wait = WebDriverWait(driver, 10)

try:
    driver.get("https://www.saucedemo.com/")
    print("Website opened successfully")

    wait.until(EC.visibility_of_element_located(
        (By.ID, "user-name"))).send_keys("standard_user")
    driver.find_element(By.ID, "password").send_keys("secret_sauce")
    driver.find_element(By.ID, "login-button").click()

    wait.until(EC.url_contains("inventory"))
    assert "inventory" in driver.current_url
    print("Login Test: PASS")

    wait.until(EC.element_to_be_clickable(
        (By.ID, "add-to-cart-sauce-labs-backpack"))).click()
    print("Product Added: PASS")

    driver.find_element(By.CLASS_NAME, "shopping_cart_link").click()
    assert "cart" in driver.current_url
    print("Cart Navigation: PASS")

    driver.find_element(By.ID, "checkout").click()
    wait.until(EC.visibility_of_element_located(
        (By.ID, "first-name"))).send_keys("Student")
    driver.find_element(By.ID, "last-name").send_keys("Testing")
    driver.find_element(By.ID, "postal-code").send_keys("380001")
    driver.find_element(By.ID, "continue").click()

    wait.until(EC.url_contains("checkout-step-two"))
    assert "checkout-step-two" in driver.current_url
    print("Multi-Page Workflow: PASS")
finally:
    driver.quit()
    print("Tutorial 11 Completed Successfully")
```

Explanation of the Code

- **Import statements —** Load Selenium's browser controls, element locators, explicit waits, and expected conditions.
- **webdriver.Chrome() —** Starts a Chrome browser controlled by Selenium.
- **driver.maximize_window() —** Maximizes the browser window.
- **WebDriverWait(driver, 10) —** Waits up to 10 seconds for a required condition.

● **driver.get(...) —** Opens the SauceDemo website.
● **By.ID / By.CLASS_NAME —** Finds webpage elements using their HTML ID or class name.
● **send_keys() —** Types text into an input field.
● **click() —** Clicks a webpage button or link.
● **assert —** Checks that the expected page is open, such as the inventory or cart page. ● **try / finally —** Runs the workflow and ensures the browser closes at the end.

Expected Output

```
Website opened successfully
Login Test: PASS
Product Added: PASS
Cart Navigation: PASS
Multi-Page Workflow: PASS
Tutorial 11 Completed Successfully
```

Screenshots

Screenshots

Use these screenshots as visual references for the practical.

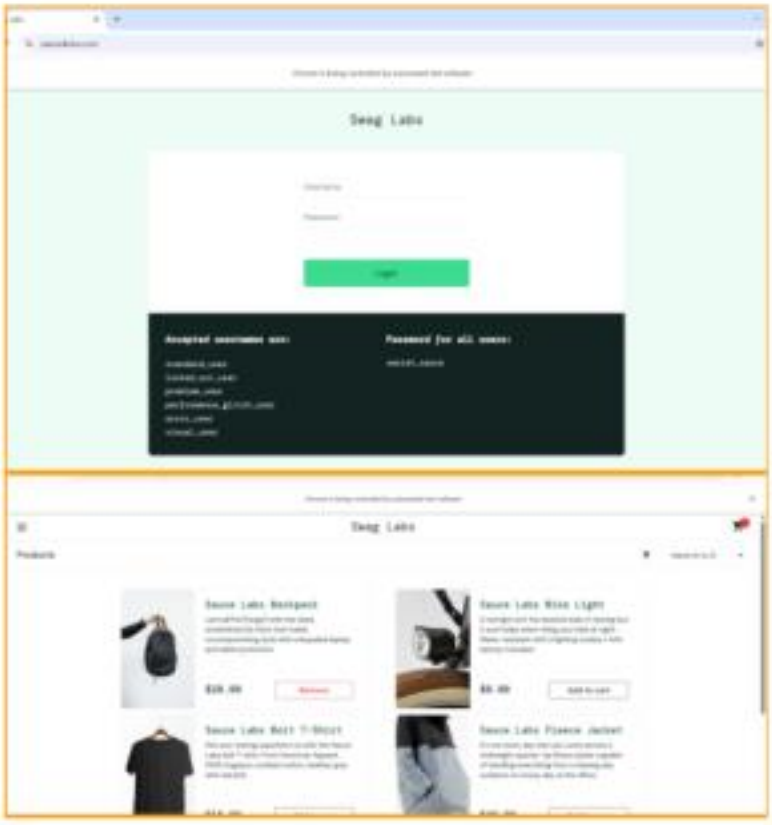
Screenshot reference – Original page 1

TUTORIAL – II

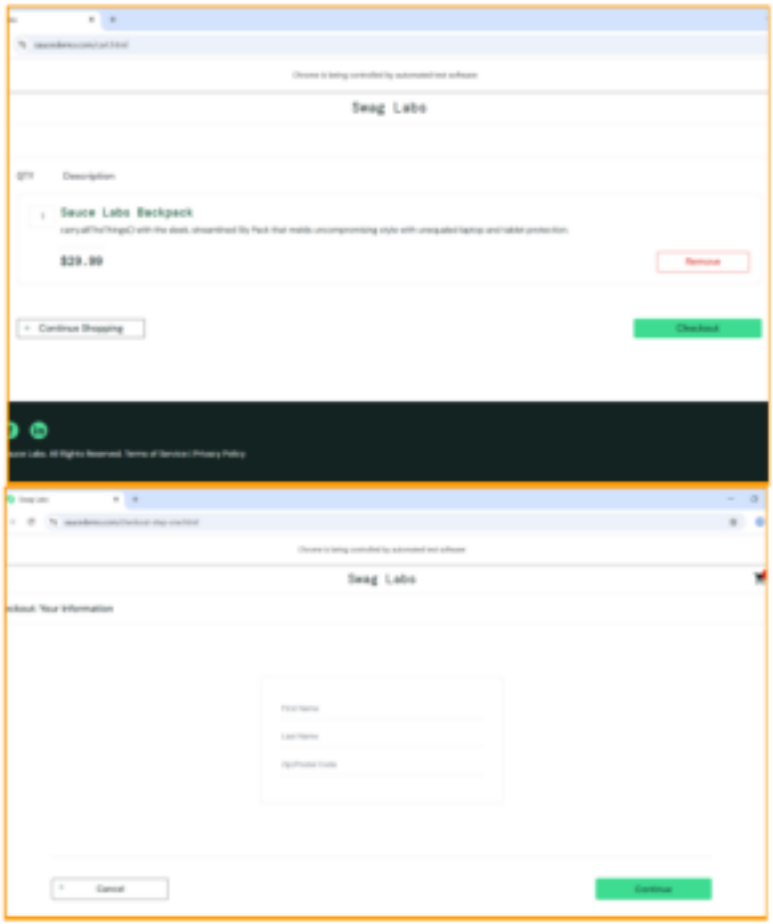
PYTHON AUTOMATION TESTING



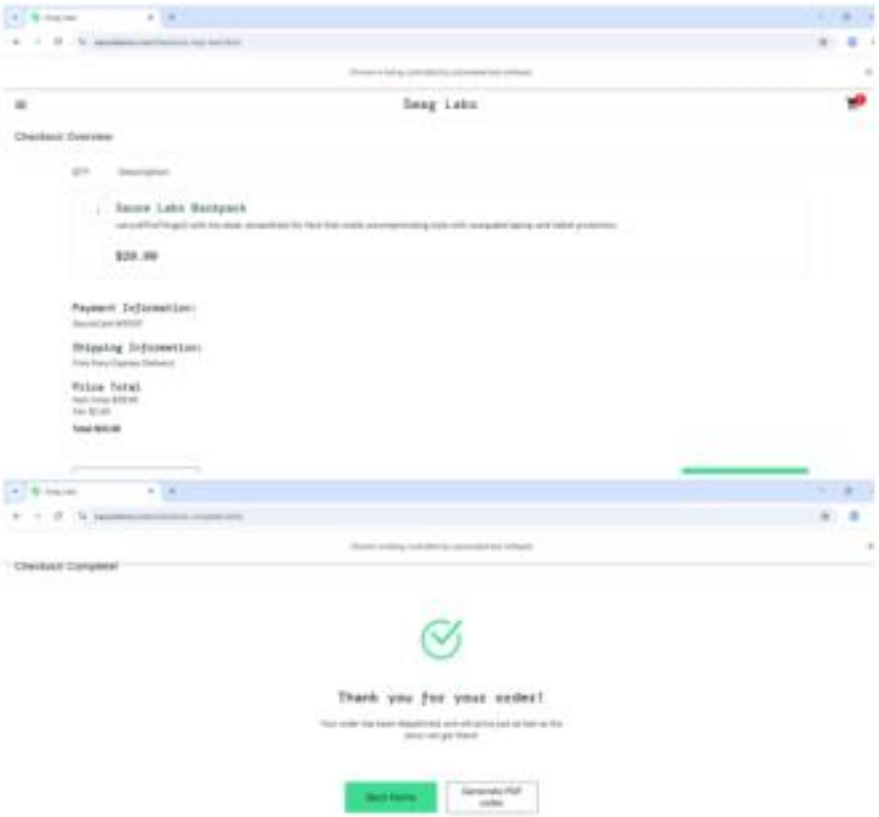
Screenshot reference – Original page 2



Screenshot reference – Original page 3



Screenshot reference – Original page 4



1. Aim

To analyze automation testing in Python for website navigation, forms, multi-page workflows, links, and regression testing scenarios.

2. Theory

What is Automation Testing?

Automation testing means using a software tool to execute test steps automatically instead of manually performing every step.

For example, manually testing login:

Conclusion

The website login, product addition, cart navigation, and checkout workflow are automated using Python and Selenium. The program verifies each major step and closes the browser after execution.